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Chamfer guidance systems and methods

Abstract

A surgical tool includes a handle and a trial section coupled to the handle by an engagement section. The trial section includes an upper surface and a lower surface. The upper surface has a shape that corresponds to an upper surface of an implant. The trial section defines a first slot extending from the upper surface to the lower surface and inwardly from a first side. A second slot extends from the upper surface to the lower surface and inwardly from a second side.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3839742	12/1973	Link	N/A	N/A
3872519	12/1974	Giannestras et al.	N/A	N/A
3886599	12/1974	Schlein	N/A	N/A
3889300	12/1974	Smith	N/A	N/A
3896502	12/1974	Lennox	N/A	N/A
3896503	12/1974	Freeman et al.	N/A	N/A
3975778	12/1975	Newton, III	N/A	N/A
3987500	12/1975	Schlein	N/A	N/A
4021864	12/1976	Waugh	N/A	N/A
4069518	12/1977	Groth, Jr. et al.	N/A	N/A
4156944	12/1978	Schreiber et al.	N/A	N/A
4166292	12/1978	Bokros	N/A	N/A
4204284	12/1979	Koeneman	N/A	N/A
4232404	12/1979	Samuelson et al.	N/A	N/A
4309778	12/1981	Buechel et al.	N/A	N/A
4470158	12/1983	Pappas et al.	N/A	N/A
4755185	12/1987	Tarr	N/A	N/A

4968316	12/1989	Hergenroeder	N/A	N/A
5041139	12/1990	Brånemark	N/A	N/A
5312412	12/1993	Whipple	N/A	N/A
5326365	12/1993	Alvine	N/A	N/A
5354300	12/1993	Goble et al.	N/A	N/A
5423825	12/1994	Levine	N/A	N/A
5476466	12/1994	Barrette et al.	N/A	N/A
5601563	12/1996	Burke et al.	N/A	N/A
5628749	12/1996	Vendrely et al.	N/A	N/A
5634927	12/1996	Houston et al.	N/A	N/A
5667511	12/1996	Vendrely et al.	N/A	N/A
5674223	12/1996	Cipolletti et al.	N/A	N/A
5735904	12/1997	Pappas	N/A	N/A
5766259	12/1997	Sammarco	623/21.18	A61B 17/15
5776200	12/1997	Johnson et al.	N/A	N/A
5817097	12/1997	Howard et al.	N/A	N/A
5824106	12/1997	Fournal	N/A	N/A
5879389	12/1998	Koshino	N/A	N/A
5885299	12/1998	Winslow et al.	N/A	N/A
5888203	12/1998	Goldberg	N/A	N/A
5897559	12/1998	Masini	N/A	N/A
5935132	12/1998	Bettuchi et al.	N/A	N/A
6002859	12/1998	DiGioia, III et al.	N/A	N/A
6033405	12/1999	Winslow et al.	N/A	N/A
6102952	12/1999	Koshino	N/A	N/A
6183519	12/2000	Bonnin et al.	N/A	N/A
6245109	12/2000	Mendes et al.	N/A	N/A
6342056	12/2001	Mac-Thiong et al.	N/A	N/A
6344043	12/2001	Pappas	N/A	N/A
6409767	12/2001	Pericéet al.	N/A	N/A
6436146	12/2001	Hassler et al.	N/A	N/A
6478800	12/2001	Fraser et al.	N/A	N/A
6520964	12/2002	Tallarida et al.	N/A	N/A
6530930	12/2002	Marino et al.	N/A	N/A
6602259	12/2002	Masini	N/A	N/A
6610067	12/2002	Tallarida et al.	N/A	N/A
6610095	12/2002	Pope et al.	N/A	N/A
6620168	12/2002	Lombardo et al.	N/A	N/A
6645215	12/2002	McGovern et al.	N/A	N/A
6663669	12/2002	Reiley	N/A	N/A
6673116	12/2003	Reiley	N/A	N/A
6679917	12/2003	Ek	N/A	N/A
6719799	12/2003	Kropf	N/A	N/A
6824567	12/2003	Tornier et al.	N/A	N/A
6852130	12/2004	Keller et al.	N/A	N/A
6860902	12/2004	Reiley	N/A	N/A
6863691	12/2004	Short et al.	N/A	N/A
6875222	12/2004	Long et al.	N/A	N/A
6875236	12/2004	Reiley	N/A	N/A

6926739	12/2004	O'Connor et al.	N/A	N/A
6939380	12/2004	Guzman	N/A	N/A
6942670	12/2004	Heldreth et al.	N/A	N/A
7001394	12/2005	Gundlapalli et al.	N/A	N/A
7011687	12/2005	Deffenbaugh et al.	N/A	N/A
7025790	12/2005	Parks et al.	N/A	N/A
7163541	12/2006	Ek	N/A	N/A
7238190	12/2006	Schon et al.	N/A	N/A
7252684	12/2006	Dearnaley	N/A	N/A
7314488	12/2007	Reiley	N/A	N/A
7323012	12/2007	Stone et al.	N/A	N/A
7476227	12/2008	Tornier et al.	N/A	N/A
7481814	12/2008	Metzger	N/A	N/A
7485147	12/2008	Papps et al.	N/A	N/A
7534246	12/2008	Reiley et al.	N/A	N/A
7534270	12/2008	Ball	N/A	N/A
7615082	12/2008	Naegerl et al.	N/A	N/A
7618421	12/2008	Axelsson, Jr. et al.	N/A	N/A
7625409	12/2008	Saltzman et al.	N/A	N/A
7641697	12/2009	Reiley	N/A	N/A
7678151	12/2009	Ek	N/A	N/A
7713305	12/2009	Ek	N/A	N/A
7717920	12/2009	Reiley	N/A	N/A
7763080	12/2009	Southworth	N/A	N/A
7803158	12/2009	Hayden	N/A	N/A
7850698	12/2009	Straszheim-Morley et al.	N/A	N/A
7896883	12/2010	Ek et al.	N/A	N/A
7896885	12/2010	Miniaci et al.	N/A	N/A
7909882	12/2010	Stinnette	N/A	N/A
7963996	12/2010	Saltzman et al.	N/A	N/A
8002841	12/2010	Hasselman	N/A	N/A
8012217	12/2010	Strzepa et al.	N/A	N/A
8034114	12/2010	Reiley	N/A	N/A
8034115	12/2010	Reiley	N/A	N/A
8048164	12/2010	Reiley	N/A	N/A
8110006	12/2011	Reiley	N/A	N/A
8114091	12/2011	Ratron et al.	N/A	N/A
8167888	12/2011	Steffensmeier	N/A	N/A
8172850	12/2011	McMinn	N/A	N/A
8177841	12/2011	Ek	N/A	N/A
8268007	12/2011	Barsoum et al.	N/A	N/A
8303667	12/2011	Younger	N/A	N/A
8313492	12/2011	Wong et al.	N/A	N/A
8317797	12/2011	Rasmussen	N/A	N/A
8323346	12/2011	Tepic	N/A	N/A
8337503	12/2011	Lian	N/A	N/A
8361159	12/2012	Ek	N/A	N/A
8475463	12/2012	Lian	N/A	N/A
8491596	12/2012	Long et al.	N/A	N/A

8808303	12/2013	Stemniski et al.	N/A	N/A
8911444	12/2013	Bailey	N/A	N/A
9492281	12/2015	Rouyer et al.	N/A	N/A
9907561	12/2017	Luna et al.	N/A	N/A
10034678	12/2017	Park et al.	N/A	N/A
10039558	12/2017	Park et al.	N/A	N/A
10149687	12/2017	McGinley et al.	N/A	N/A
10182832	12/2018	Saltzman et al.	N/A	N/A
10206688	12/2018	Park et al.	N/A	N/A
10940012	12/2020	Sander et al.	N/A	N/A
2002/0068977	12/2001	Jackson	N/A	N/A
2002/0082607	12/2001	Heldreth et al.	N/A	N/A
2002/0133164	12/2001	Williamson	N/A	N/A
2002/0173853	12/2001	Corl, III et al.	N/A	N/A
2003/0208280	12/2002	Tohidi	N/A	N/A
2003/0236522	12/2002	Long et al.	N/A	N/A
2004/0030399	12/2003	Asencio	N/A	N/A
2004/0039394	12/2003	Conti et al.	N/A	N/A
2004/0068322	12/2003	Ferree	N/A	N/A
2004/0167631	12/2003	Luchesi et al.	N/A	N/A
2004/0186585	12/2003	Feiwell	N/A	N/A
2004/0216259	12/2003	Ponziani	N/A	N/A
2004/0236431	12/2003	Sekel	N/A	N/A
2005/0004676	12/2004	Schon et al.	N/A	N/A
2005/0165408	12/2004	Puno et al.	N/A	N/A
2005/0192674	12/2004	Ferree	N/A	N/A
2005/0273113	12/2004	Kuczynski	N/A	N/A
2006/0009857	12/2005	Gibbs et al.	N/A	N/A
2006/0020345	12/2005	O'Connor et al.	N/A	N/A
2006/0036257	12/2005	Steffensmeier	N/A	N/A
2006/0116679	12/2005	Lutz et al.	N/A	N/A
2006/0142870	12/2005	Robinson et al.	N/A	N/A
2006/0235541	12/2005	Hodorek	N/A	N/A
2006/0247788	12/2005	Ross	N/A	N/A
2007/0038303	12/2006	Myerson et al.	N/A	N/A
2007/0100346	12/2006	Wyss et al.	N/A	N/A
2007/0112431	12/2006	Kofoed	N/A	N/A
2007/0162025	12/2006	Tornier et al.	N/A	N/A
2007/0173944	12/2006	Keller et al.	N/A	N/A
2007/0173947	12/2006	Ratron	N/A	N/A
2007/0213830	12/2006	Ammann et al.	N/A	N/A
2007/0233129	12/2006	Bertagnoli et al.	N/A	N/A
2007/0276400	12/2006	Moore et al.	N/A	N/A
2007/0288030	12/2006	Metzger et al.	N/A	N/A
2008/0015602	12/2007	Axelson	N/A	N/A
2008/0097617	12/2007	Fellinger et al.	N/A	N/A
2008/0103603	12/2007	Hintermann	N/A	N/A
2008/0109081	12/2007	Bao et al.	N/A	N/A
2008/0195233	12/2007	Ferrari et al.	N/A	N/A
2008/0215156	12/2007	Duggal et al.	N/A	N/A

2008/0287954	12/2007	Kunz et al.	N/A	N/A
2008/0312745	12/2007	Keller et al.	N/A	N/A
2009/0024131	12/2008	Metzger et al.	N/A	N/A
2009/0043309	12/2008	Rasmussen	N/A	N/A
2009/0043310	12/2008	Rasmussen	N/A	N/A
2009/0054992	12/2008	Landes et al.	N/A	N/A
2009/0082875	12/2008	Long	N/A	N/A
2009/0088758	12/2008	Bennett	606/82	A61B 17/155
2009/0105767	12/2008	Reiley	N/A	N/A
2009/0105840	12/2008	Reiley	N/A	N/A
2009/0182433	12/2008	Reiley et al.	N/A	N/A
2009/0198341	12/2008	Choi et al.	N/A	N/A
2009/0234360	12/2008	Alexander	N/A	N/A
2009/0276052	12/2008	Regala et al.	N/A	N/A
2010/0010493	12/2009	Dower	N/A	N/A
2010/0023066	12/2009	Long et al.	N/A	N/A
2010/0023126	12/2009	Grotz	N/A	N/A
2010/0057216	12/2009	Gannoe et al.	N/A	N/A
2010/0069910	12/2009	Hasselman	N/A	N/A
2010/0198355	12/2009	Kofoed et al.	N/A	N/A
2010/0212138	12/2009	Carroll et al.	N/A	N/A
2010/0241237	12/2009	Pappas	N/A	N/A
2010/0305572	12/2009	Saltzman et al.	N/A	N/A
2010/0318088	12/2009	Warne et al.	N/A	N/A
2010/0331984	12/2009	Barsoum et al.	N/A	N/A
2011/0029090	12/2010	Zannis et al.	N/A	N/A
2011/0035018	12/2010	Deffenbaugh et al.	N/A	N/A
2011/0035019	12/2010	Goswami et al.	N/A	N/A
2011/0071645	12/2010	Bojarski et al.	N/A	N/A
2011/0106268	12/2010	Deffenbaugh et al.	N/A	N/A
2011/0112542	12/2010	Gross	N/A	N/A
2011/0125200	12/2010	Hanson et al.	N/A	N/A
2011/0125275	12/2010	Lipman et al.	N/A	N/A
2011/0125284	12/2010	Gabbrielli et al.	N/A	N/A
2011/0152868	12/2010	Kourtis et al.	N/A	N/A
2011/0152869	12/2010	Ek et al.	N/A	N/A
2011/0166608	12/2010	Duggal et al.	N/A	N/A
2011/0190829	12/2010	Duggal et al.	N/A	N/A
2011/0218542	12/2010	Lian	N/A	N/A
2011/0245835	12/2010	Dodd et al.	N/A	N/A
2011/0253151	12/2010	Tochigi et al.	N/A	N/A
2011/0276052	12/2010	Hasselman	N/A	N/A
2011/0295380	12/2010	Long	N/A	N/A
2012/0010718	12/2011	Still	N/A	N/A
2012/0046753	12/2011	Cook et al.	N/A	N/A
2012/0053591	12/2011	Haines et al.	N/A	N/A
2012/0053644	12/2011	Landry et al.	N/A	N/A
2012/0083789	12/2011	Blakemore et al.	N/A	N/A
2012/0109131	12/2011	Vasarhelyi et al.	N/A	N/A

2012/0109326	12/2011	Perler	N/A	N/A
2012/0130376	12/2011	Loring et al.	N/A	N/A
2012/0136443	12/2011	Wenzel	N/A	N/A
2012/0185057	12/2011	Abidi et al.	N/A	N/A
2012/0191210	12/2011	Ratron et al.	N/A	N/A
2012/0239045	12/2011	Li	N/A	N/A
2012/0245701	12/2011	Zak et al.	N/A	N/A
2012/0271430	12/2011	Arnett et al.	N/A	N/A
2012/0277745	12/2011	Lizee	N/A	N/A
2013/0041473	12/2012	Rouyer et al.	N/A	N/A
2013/0116797	12/2012	Coulange et al.	N/A	N/A
2014/0107655	12/2013	Song	606/88	A61B 17/157
2014/0276853	12/2013	Long	606/87	A61F 2/4202
2014/0309640	12/2013	Smith et al.	N/A	N/A
2015/0045801	12/2014	Axelson et al.	N/A	N/A
2016/0135815	12/2015	Loring et al.	N/A	N/A
2018/0177511	12/2017	Luna et al.	N/A	N/A
2018/0263639	12/2017	McGinley et al.	N/A	N/A
2019/0059917	12/2018	Saltzman	N/A	N/A
2019/0059918	12/2018	Saltzman et al.	N/A	N/A
2019/0133612	12/2018	McGinley et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2836651	12/2015	CA	N/A
101790353	12/2009	CN	N/A
2967697	12/2017	EP	N/A
3354233	12/2018	EP	N/A
2480846	12/2010	GB	N/A
H11-500035	12/1998	JP	N/A
2006150055	12/2005	JP	N/A
2007518453	12/2006	JP	N/A
2007519477	12/2006	JP	N/A
2007536011	12/2006	JP	N/A
2011526189	12/2010	JP	N/A
2012518517	12/2011	JP	N/A
2013500810	12/2012	JP	N/A
2013511358	12/2012	JP	N/A
2014131738	12/2013	JP	N/A
WO 9625106	12/1995	WO	N/A
WO 0166021	12/2000	WO	N/A
WO 2005011523	12/2004	WO	N/A
WO 2006022923	12/2005	WO	N/A
WO 2006023824	12/2005	WO	N/A
WO 2006099270	12/2005	WO	N/A
WO 2007084846	12/2006	WO	N/A
WO 2009158522	12/2008	WO	N/A

WO 2010099142	12/2009	WO	N/A
WO 2011015863	12/2010	WO	N/A
WO 2011063281	12/2010	WO	N/A
WO 2011151657	12/2010	WO	N/A
WO 2012088036	12/2011	WO	N/A
WO 2012116089	12/2011	WO	N/A
WO 2016039762	12/2015	WO	N/A
2019091537	12/2018	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report issued in connection with corresponding European Patent Application No. 21788454.3, Mar. 1, 2024, 10 pages. cited by applicant

Search report issued for European patent application No. 13198280 dated Feb. 5, 2014. cited by applicant

International Search Report for International patent application No. PCT/US2014/027448 dated Jul. 7, 2014. cited by applicant

International Preliminary Report on Patentability issued for International patent application No. PCT/US2014/027448, Sep. 15, 2015, 8 pages. cited by applicant

Partial European Search Report issued in connection with European patent application No. 14768333.8, Oct. 26, 2016, 6 pages. cited by applicant

Patent Examination Report No. 1 issued in connection with Australian patent application No. 2015202080, Jul. 5, 2016, 4 pages. cited by applicant

First Office Action issued for Japanese patent application No. 2016-117842, Sep. 12, 2017, 5 pages. cited by applicant

First Office Action issued in connection with corresponding Japanese Patent Application No. 2020-016447, Apr. 6, 2021, 4 pages. cited by applicant

Office Action in corresponding Canadian Patent Application No. 2,904,652, Jun. 2, 2020, 6 pages. cited by applicant

First Examination Report issued in corresponding Australian Patent Application No. 2019213412, Sep. 3, 2020, 5 pages. cited by applicant

First Office Action in corresponding Canadian Patent Application No. 2,904,652, Jan. 28, 2020, 5 pages. cited by applicant

Final Office Action issued in connection with corresponding Japanese Patent Application No. 206-502443, May 15, 2018, 3 pages. cited by applicant

Extended European Search Report issued in connection with corresponding European Patent Application No. 18160378.8, Jun. 29, 2018, 7 pages. cited by applicant

Second Office Action issued in connection with corresponding Chinese Patent Application No. 2018071101785100, dated Jul. 16, 2016, 6 pages. cited by applicant

First Office Action in corresponding Japanese Patent Application No. 2018-178853, Sep. 3, 2018, 3 pages. cited by applicant

Examination Report No. 1 issued in connection with corresponding Australian Patent Application No. 20182000073, Dec. 24, 2018, 3 pages. cited by applicant

First Office Action issued in connection with corresponding Japanese Patent Application No. 2018-092289, Mar. 5, 2019, 2 pages. cited by applicant

Extended European Search Report and Opinion in connection with European Patent Application No. 14768333.8, dated Jan. 30, 2017, 10 pages. cited by applicant

First Office Action issued in connection with Chinese Patent Application No. 2017800899442 dated Apr. 6, 2022, 8 pages. cited by applicant

International Search Report and Written Opinion issued in connection with International Patent Application No. PCT/US2021/025873, Sep. 2, 2021. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage Application, filed under 35 U.S.C. 371, of International Patent Application No. PCT/US2021/025873, filed on Apr. 6, 2021, which claims priority to U.S. Provisional Patent Application No. 63/010,895, filed on Apr. 16, 2020, the entire contents of which are incorporated herein by reference.

INCORPORATION BY REFERENCE

(1) This application also incorporates by reference in its entirety U.S. Pat. No. 9,907,561, entitled “Ankle Replacement System and Method,” which issued from U.S. patent application Ser. No. 14/446,921 on Mar. 6, 2018.

FIELD OF DISCLOSURE

(2) The disclosure relates to surgical devices and methods. More specifically, the disclosure relates to system and methods for total ankle replacement.

BACKGROUND

(3) U.S. Pat. No. 9,907,561 (the “561 Patent”), which patent has been incorporated by reference in its entirety as set forth above, discloses various tools for performing total ankle arthroplasty. The procedure for preparing the talus in the '561 Patent requires numerous steps and surgical tools. Accordingly, systems and methods for talus preparation using fewer components or steps would be beneficial, as it would reduce the length of the surgeries and manufacturing costs.

SUMMARY

(4) In some embodiments, a surgical tool includes a handle and a trial section coupled to the handle by an engagement section. The trial section includes an upper surface and a lower surface. The upper surface has a shape that corresponds to an upper surface of an implant. The trial section defines a first slot extending from the upper surface to the lower surface and inwardly from a first side. A second slot extends from the upper surface to the lower surface and inwardly from a second side.

(5) A tool includes a base component and arm component. The base component has a first surface for engaging a bone, a second surface disposed on an opposite side of the base component with respect to the first surface, and a third surface extending between the first surface and the second surface. The first surface defines a first slot that extends inwardly from a first side of the base component to a second side of the base component. The base component defines a second slot extending inwardly from the third surface and being in communication with the first slot, and the second surface defining a passageway. The arm component has a base from which a peg extends. The peg is size and configured to be at least partially received within the passageway defined by the base component for coupling the arm component to the base component. The arm component defines an opening sized and configured to receive a first cutting tool therein.

(6) In some embodiments, a tool includes a body extending from a first end to a second end. The body includes a first surface disposed adjacent to the first end that is adapted to contact a bone. The surface defines a first slot extending from a first side of the body to a second side of the body. The body includes an extension having a second surface disposed at the second end. The extension defines a second slot sized and configured to receive a cutting tool therein. The second slot has a longitudinal axis that extends in a widthwise direction across the body. The body defines a third

slot along a length of the body between the first end and the second end, and the third slot extending through the body such that the third slot is in communication with the first slot.

(7) In some embodiments, a tool includes a base component and a guide component for coupling to the base component. The base component defines a slot, a first pair of holes located on a first side of the slot, and a second pair of holes located on a second side of the slot. The guide component includes a pair of pegs extending from a first surface. The pair of pegs sized and arranged to be received in the first pair of holes and the second pair of holes of the base component. The guide component defines a first slot and a second slot that extend parallel to one another. The first slot defined by the guide is aligned with the slot defined by the base component when the pair of pegs are received within the first pair of holes, and the second slot defined by the guide is aligned with the slot defined by the base component when the pair of pegs are received within the second pair of holes.

(8) In some embodiments, a tool includes a base component and an arm component for coupling to the base component. The base component extends from a first end to a second end and defines at least one hole adjacent to the first end and including a beam disposed at the second end. The beam includes at least one projection extending along its length. The arm component includes a coupling end defining a recess for receiving at least a portion of the beam and the at least one projection to couple the arm component to the base component. The arm component defines at least one hole sized and configured to receive a cutting tool therethrough.

(9) In some embodiments, a tool includes locator guide. The locator guide extends from a first end to a second end and defines at least one hole sized and configured to receive a fixation element. The locator guide contains features that interact with mating features of a cutting guide.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an isometric view of one example of a multi-purpose tool in accordance with some embodiments;
- (2) FIG. 2 is a top side plan view of the multi-purpose tool illustrated in FIG. 1 in accordance with some embodiments;
- (3) FIG. 3 is a bottom side plan view of the multi-purpose tool illustrated in FIG. 1 in accordance with some embodiments;
- (4) FIG. 4 is a side view of the multi-purpose tool illustrated in FIG. 1 in accordance with some embodiments;
- (5) FIG. 5 is an end view of the multi-purpose tool illustrated in FIG. 1 in accordance with some embodiments;
- (6) FIG. 6 is an isometric view of one example of a resection guide in accordance with some embodiments;
- (7) FIG. 7 is an isometric view of a base component of the resection guide illustrated in FIG. 6 in accordance with some embodiments;
- (8) FIG. 8 is a side view of the base component illustrated in FIG. 7 in accordance with some embodiments;
- (9) FIG. 9 is a rear side view of the base component illustrated in FIG. 7 in accordance with some embodiments;
- (10) FIG. 10 is a front side view of the base component illustrated in FIG. 7 in accordance with some embodiments;
- (11) FIG. 11 is a top side view of the base component illustrated in FIG. 7 in accordance with some embodiments;
- (12) FIG. 12 is a bottom side view of the base component illustrated in FIG. 7 in accordance with

some embodiments;

(13) FIG. 13 is an isometric view of the arm component of the resection guide illustrated in FIG. 6 in accordance with some embodiments;

(14) FIG. 14 is an isometric view of a base of the arm component illustrated in FIG. 13 in accordance with some embodiments;

(15) FIG. 15 is a front side view of the base illustrated in FIG. 14 in accordance with some embodiments;

(16) FIG. 16 is a section view of the base taken along line 16-16 in FIG. 15 in accordance with some embodiments;

(17) FIG. 17 is an isometric view of an arm of the arm component illustrated in FIG. 13 in accordance with some embodiments;

(18) FIG. 18 is a top side view of the arm illustrated in FIG. 17 in accordance with some embodiments;

(19) FIG. 19 is a bottom side view of the arm illustrated in FIG. 17 in accordance with some embodiments;

(20) FIG. 20 is an isometric view of one example of a dowel that may be used in connection with the resection guide illustrated in FIG. 6 in accordance with some embodiments;

(21) FIG. 21 is a front side view of the dowel illustrated in FIG. 20 in accordance with some embodiments;

(22) FIG. 22 is an isometric view of one example of a cutting tool that may be used in connection with the resection guide illustrated in FIG. 6 in accordance with some embodiments;

(23) FIG. 23 illustrates a side view of the cutting tool illustrated in FIG. 22 in accordance with some embodiments;

(24) FIG. 24 is a top side view of the cutting tool illustrated in FIG. 22 in accordance with some embodiments;

(25) FIG. 25 is a bottom side view of the cutting tool illustrated in FIG. 22 in accordance with some embodiments;

(26) FIG. 26 is an isometric view of one example of a resection guide in accordance with some embodiments;

(27) FIG. 27 is a front end view of the resection guide illustrated in FIG. 26 in accordance with some embodiments;

(28) FIG. 28 is a rear end view of the resection guide illustrated in FIG. 26 in accordance with some embodiments;

(29) FIG. 29 is a bottom side view of the resection guide illustrated in FIG. 26 in accordance with some embodiments;

(30) FIG. 30 is a top side view of the resection guide illustrated in FIG. 26 in accordance with some embodiments;

(31) FIG. 31 is a side view of the resection guide illustrated in FIG. 26 in accordance with some embodiments;

(32) FIG. 32 is an isometric view of one example of a fixation element in accordance with some embodiments;

(33) FIG. 33 is a side view of the fixation element illustrated in FIG. 32 in accordance with some embodiments;

(34) FIG. 34 is an isometric view of one example of a reaming cutting tool in accordance with some embodiments;

(35) FIG. 35 a side view of the reaming cutting tool illustrated in FIG. 34 in accordance with some embodiments;

(36) FIG. 36 is another side view of the reaming cutting tool illustrated in FIG. 34 in accordance with some embodiments;

(37) FIG. 37 is an end view of the reaming cutting tool illustrated in FIG. 34 in accordance with

some embodiments;

(38) FIG. **38** is an isometric view of one example of an angled cutting tool in accordance with some embodiments;

(39) FIG. **39** is a side view of the angled cutting tool illustrated in FIG. **38** in accordance with some embodiments;

(40) FIG. **40** is a side view of the angled cutting tool illustrated in FIG. **38** in accordance with some embodiments;

(41) FIG. **41** illustrates one example of the resection guide illustrated in FIGS. **26-31** being secured to bone and guiding cutting tools in accordance with some embodiments;

(42) FIG. **42** is an isometric view of one example of a cutting guide in accordance with some embodiments;

(43) FIG. **43** is a side view of the cutting guide illustrated in FIG. **42** in accordance with some embodiments;

(44) FIG. **44** is a top side view of the cutting guide illustrated in FIG. **42** in accordance with some embodiments;

(45) FIG. **45** is a bottom side view of the cutting guide illustrated in FIG. **42** in accordance with some embodiments;

(46) FIG. **46** is a front side view of the cutting guide illustrated in FIG. **42** in accordance with some embodiments;

(47) FIG. **47** is a rear side view of the cutting guide illustrated in FIG. **42** in accordance with some embodiments;

(48) FIG. **48** is an isometric view of a cutting guide in accordance with some embodiments;

(49) FIG. **49** is a rear view of the cutting guide illustrated in FIG. **48** in accordance with some embodiments;

(50) FIG. **50** is a front side view of the cutting guide illustrated in FIG. **48** in accordance with some embodiments;

(51) FIG. **51** is a side view of the cutting guide illustrated in FIG. **48** in accordance with some embodiments;

(52) FIG. **52** is a top side view of the cutting guide illustrated in FIG. **48** in accordance with some embodiments;

(53) FIG. **53** is a bottom side view of the cutting guide illustrated in FIG. **48** in accordance with some embodiments;

(54) FIG. **54** illustrates one example of the cutting guide illustrated in FIG. **48** coupled to bone in accordance with some embodiments;

(55) FIG. **55** illustrates one example of a cutting guide and guiding a cutting tool in accordance with some embodiments;

(56) FIG. **56** is an isometric view of one example of a base component of the cutting guide illustrated in FIG. **55** in accordance with some embodiments;

(57) FIG. **57** is a top side plan view of the base component illustrated in FIG. **56** in accordance with some embodiments;

(58) FIG. **58** is a bottom side plan view of the base component illustrated in FIG. **56** in accordance with some embodiments;

(59) FIG. **59** is a front side view of the base component illustrated in FIG. **56** in accordance with some embodiments;

(60) FIG. **60** is a side view of the base component illustrated in FIG. **56** in accordance with some embodiments;

(61) FIG. **61** is an isometric view of an arm component of the cutting guide illustrated in FIG. **55** in accordance with some embodiments;

(62) FIG. **62** is a side view of the arm component illustrated in FIG. **61** in accordance with some embodiments;

(63) FIG. **63** is a top side plan view of the arm component illustrated in FIG. **61** in accordance with some embodiments;

(64) FIG. **64** is a view of the coupling end of the arm component illustrated in FIG. **61** in accordance with some embodiments;

(65) FIG. **65** is an isometric view of one example of a cutting tool in accordance with some embodiments;

(66) FIG. **66** is a side view of the cutting tool illustrated in FIG. **65** in accordance with some embodiments;

(67) FIG. **67** is a top side view of the cutting tool illustrated in FIG. **65** in accordance with some embodiments;

(68) FIG. **68** is an isometric view of one example of a nut in accordance with some embodiments;

(69) FIG. **69** is a top side plan view of the nut illustrated in FIG. **68** in accordance with some embodiments;

(70) FIG. **70** is a side view of the nut illustrated in FIG. **68** in accordance with some embodiments;

(71) FIG. **71** is an isometric view of another example of an arm component of the cutting guide illustrated in FIG. **55** in accordance with some embodiments;

(72) FIG. **72** is a side view of the arm component illustrated in FIG. **71** in accordance with some embodiments;

(73) FIG. **73** is a top side view of the arm component illustrated in FIG. **71** in accordance with some embodiments;

(74) FIG. **74** is a bottom side view of the arm component illustrated in FIG. **71** in accordance with some embodiments;

(75) FIG. **75** is an end view of the arm component illustrated in FIG. **71** in accordance with some embodiments;

(76) FIG. **76** is an isometric view of one example of a cutting tool in accordance with some embodiments;

(77) FIG. **77** is a top side view of the cutting tool illustrated in FIG. **76** in accordance with some embodiments;

(78) FIG. **78** is a side view of the cutting tool illustrated in FIG. **76** in accordance with some embodiments;

(79) FIG. **79** is another side view of the cutting tool illustrated in FIG. **76** in accordance with some embodiments;

(80) FIG. **80** illustrates one example of a cutting guide being used to perform a cut in accordance with some embodiments;

(81) FIG. **81** illustrates one example of a cutting guide being used to perform another cut in accordance with some embodiments;

(82) FIG. **82** is an isometric view of another example of a cutting guide base in accordance with some embodiments;

(83) FIG. **83** is a side view of the cutting guide base illustrated in FIG. **82** in accordance with some embodiments;

(84) FIG. **84** is a front side view of the cutting guide base illustrated in FIG. **82** in accordance with some embodiments;

(85) FIG. **85** is a rear side view of the cutting guide base illustrated in FIG. **82** in accordance with some embodiments;

(86) FIG. **86** is a top side view of the cutting guide base illustrated in FIG. **82** in accordance with some embodiments;

(87) FIG. **87** is a bottom side view of the cutting guide base illustrated in FIG. **82** in accordance with some embodiments;

(88) FIG. **88** is an isometric view of one example of a locator guide in accordance with some embodiments;

- (89) FIG. **89** is a top side view of the locator guide illustrated in FIG. **88** in accordance with some embodiments;
- (90) FIG. **90** is a bottom side view of the locator guide illustrated in FIG. **88** in accordance with some embodiments;
- (91) FIG. **91** is a front side view of the locator guide illustrated in FIG. **88** in accordance with some embodiments;
- (92) FIG. **92** is a rear side view of the locator guide illustrated in FIG. **88** in accordance with some embodiments;
- (93) FIG. **93** is a side view of the locator guide illustrated in FIG. **88** in accordance with some embodiments;
- (94) FIG. **94** is an isometric view of a cutting guide in accordance with some embodiments;
- (95) FIG. **95** is a side view of the cutting guide illustrated in FIG. **94** in accordance with some embodiments;
- (96) FIG. **96** is a front side view of the cutting guide illustrated in FIG. **94** in accordance with some embodiments;
- (97) FIG. **97** is a rear side view of the cutting guide illustrated in FIG. **94** in accordance with some embodiments;
- (98) FIG. **98** is a top side view of the cutting guide illustrated in FIG. **94** in accordance with some embodiments;
- (99) FIG. **99** is a bottom side view of the cutting guide illustrated in FIG. **94** in accordance with some embodiments;
- (100) FIG. **100** illustrates one example of a posterior cutting guide being coupled to bone and to a cutting guide base in accordance with some embodiments;
- (101) FIG. **101** is an isometric view of another example of a cutting guide in accordance with some embodiments;
- (102) FIG. **102** is a side view of the cutting guide illustrated in FIG. **101** in accordance with some embodiments;
- (103) FIG. **103** is a front side view of the cutting guide illustrated in FIG. **101** in accordance with some embodiments;
- (104) FIG. **104** is a rear side view of the cutting guide illustrated in FIG. **101** in accordance with some embodiments;
- (105) FIG. **105** is a top side view of the cutting guide illustrated in FIG. **101** in accordance with some embodiments;
- (106) FIG. **106** is a bottom side view of the cutting guide illustrated in FIG. **101** in accordance with some embodiments;
- (107) FIG. **107** illustrates one example of a posterior cutting guide being coupled to bone and to a cutting guide base in accordance with some embodiments;
- (108) FIG. **108** is an isometric view another example of a cutting guide in accordance with some embodiments;
- (109) FIG. **109** is a rear side view of the cutting guide illustrated in FIG. **108** in accordance with some embodiments;
- (110) FIG. **110** is a front side view of the cutting guide illustrated in FIG. **108** in accordance with some embodiments;
- (111) FIG. **111** is a top side view of the cutting guide illustrated in FIG. **108** in accordance with some embodiments;
- (112) FIG. **112** is a bottom side view of the cutting guide illustrated in FIG. **108** in accordance with some embodiments;
- (113) FIG. **113** is a side view of the cutting guide illustrated in FIG. **108** in accordance with some embodiments;
- (114) FIG. **114** is an isometric view of another example of a cutting guide that may be coupled to

the cutting guide illustrated in FIGS. **108-113** in accordance with some embodiments;
(115) FIG. **115** is a front side view of the cutting guide illustrated in FIG. **114** in accordance with some embodiments
(116) FIG. **116** is a rear side view of the cutting guide illustrated in FIG. **114** in accordance with some embodiments;
(117) FIG. **117** is a side view of the cutting guide illustrated in FIG. **114** in accordance with some embodiments;
(118) FIG. **118** is a top side view of the cutting guide illustrated in FIG. **114** in accordance with some embodiments;
(119) FIG. **119** is a bottom side view of the cutting guide illustrated in FIG. **114** in accordance with some embodiments;
(120) FIG. **120** is an isometric view of the guides illustrated in FIGS. **108-113** and **114-119** coupled together in accordance with some embodiments;
(121) FIG. **121** is an isometric view of a base component in accordance with some embodiments;
(122) FIG. **122** is a top side view of the base component illustrated in FIG. **121** in accordance with some embodiments;
(123) FIG. **123** is a bottom side view of the base component illustrated in FIG. **121** in accordance with some embodiments;
(124) FIG. **124** is a rear side view of the base component illustrated in FIG. **121** in accordance with some embodiments;
(125) FIG. **125** is a front side view of the base component illustrated in FIG. **121** in accordance with some embodiments;
(126) FIG. **126** is a side view of the base component illustrated in FIG. **121** in accordance with some embodiments;
(127) FIG. **127** is an isometric view of a cutting guide that may be coupled to the base component illustrated in FIGS. **121-126** in accordance with some embodiments;
(128) FIG. **128** is a front side view of the cutting guide illustrated in FIG. **127** in accordance with some embodiments;
(129) FIG. **129** is a rear side view of the cutting guide illustrated in FIG. **127** in accordance with some embodiments;
(130) FIG. **130** is a top side view of the cutting guide illustrated in FIG. **127** in accordance with some embodiments;
(131) FIG. **131** is a bottom side view of the cutting guide illustrated in FIG. **127** in accordance with some embodiments; and
(132) FIGS. **132** and **133** are opposing side views of the cutting guide illustrated in FIG. **126** in accordance with some embodiments.

DETAILED DESCRIPTION

(133) This description of the exemplary embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description.

(134) The inventors have identified certain improvements to various components disclosed in the '561 Patent that advantageously enables the procedures to be performed using fewer components, which also enables the procedures to be performed faster. Although the disclosed systems and methods are described in connection with the systems and methods disclosed in the '561 Patent, a person of ordinary skill in the art will understand that the disclosure is not so limited as the disclosed systems and methods may be used in a variety of different surgical procedures and with other systems or components.

(135) Multi-Purpose Talar Dome Trial

(136) FIGS. **1-5** illustrate one example of an improved multi-purpose talar dome trial in accordance with some embodiments. As described below, in addition to being used as a trial, trial **100** may also be used as a cutting guide for making bony cuts, including chamfer cuts to the talus.

(137) The talar dome trial **100** illustrated in FIG. 1 includes a handle section **102**, an engagement section **104**, and a trial section **106**. In some embodiments, handle section **102** has an elongate shape extending from a first end **108** of the trial **100** to the engagement section **104**. As best seen in FIGS. 1 and 2, handle section **102** may define a hole **110** adjacent to first end **108** of trial **100**.

(138) In some embodiments, engagement section **104** includes a body **112** having a thickness that is greater than a thickness of the handle section **102** as best seen in FIG. 4. However, one of ordinary skill in the art will understand that the thickness of handle section **102** may be varied such that the thickness of engagement section **104** may be equal to or less than the thickness of handle section **102**. Body **112** may have a generally rectangular shape with a sloped leading side **114**, as best seen in FIGS. 1 and 4, and define a pair of spaced apart holes **116-1**, **116-2** (collectively, “holes **116**”). In some embodiments, holes **116** extend through the entirety of body **112** and may have central axes that are parallel to a plane defined by side **114** of body **112**.

(139) Trial section **106** has at least one convex upper surface **122** that corresponds to the size and shape of a prosthetic talar dome of an ankle replacement system. As best seen in FIGS. 1-2, a pair of spaced apart slots **124-1**, **124-2** (collectively, “slots **124**”) are defined by the trial section **106** such that the slots **124** extend from the upper surface **122** to the bottom surface **126**. In some embodiments, the angle at which the slots extend at an angle through trial section **106** is parallel to an angle at which the holes **116** extend through engagement section **104**.

(140) As best seen in FIG. 3, trial section **106** further defines a pair of angle slots **128-1**, **128-2** (collectively, “slots **128**”) that inwardly extend from the bottom surface **126** of trial **100** from a first side **132** (e.g., a lateral side) to a second side (e.g., a medial side) **134**. A portion of slots **128** are aligned and communicate with slots **124** as best seen in FIG. 3. As described in greater detail below, slots **128** may be used to by a surgeon or other practitioner as a visual indicator under fluoroscopy for checking where the posterior chamfer cut will be made.

(141) In use, the multi-purpose trial **100** may be inserted into a resected joint space formed between the inferior portion of a tibia and the superior portion of the talus. The trial **100** may be inserted into the resected joint space along with a poly trial insert, such as poly trial insert **230** disclosed in the '561 Patent. The location of the trial **100** may be checked using sagittal plane fluoroscopy to confirm that the posterior portion of the trial section **106** of trial **100** rests on the posterior portion of the talus (e.g., to establish congruence between the talus and the implant). Slots **128**, which are visible under fluoroscopy, also may be used to visualize to location at which the posterior chamfer cuts will be made (i.e., extending along an axis defined by the slots **128**).

(142) When the desired positioning of trial **100** is achieved, the trial **100** may be secured to the talus by inserting one or more pins, k-wires, or other fixation elements through holes **116**. Other instruments, such as a poly insert, may be removed from the resected joint space leaving behind the trial **100**. Posterior chamfer cuts may then be made using the trial **100** as a cutting guide. For example, with trial **100** secured to the talus, a surgeon or other practitioner may make bony cuts using a bone saw by inserting the bone saw into slots **124**, **128**. Advantageously, the trial **100** enables saw cuts to be made through the medial and lateral edges of the talus. Once the talar cut has been made, the trial **100** may be removed from its engagement leaving the pins within the talus.

(143) Anterior/Posterior Resection Guide

(144) FIGS. 6-21 illustrate one example of a chamfer guide **200** in accordance with some embodiments. Chamfer guide **200** includes a base component **202** that supports an arm component **240**. The body **204** of base component **202** includes a first leg **206** and a second leg **208** that extends away from first leg **206** at an angle. Leg **206** includes a planar bottom surface **210** that defines a slot **212**, which extends from a first side **214** to a second side **216**.

(145) As best seen in FIGS. 7, 9, and 11, body **204** defines a pair of spaced apart recesses **218-1**, **218-2** (collectively, “recesses **218**”) each defining a respective hole **220-1**, **220-2** (collectively, “holes **220**”). Holes **220** extend through body **204**, extending through bottom surface **210**, and being sized and configured to receive a pin, k-wire, or other fixation element therein. In some

embodiments, holes **220** extend through body **204** parallel to one another; however, one of ordinary skill in the art will understand that holes **220** may extend through body such that they are not disposed parallel to one another. Leg **208** includes a planar upper surface **222** defining a passageway **224**. As best seen in FIG. **11**, in some embodiments, passageway **224** extends from side **214** across body **204** terminating at end wall **226**. Rear surface **228** defines a slot **230** that extends at an angle through body **204** to bottom surface **210** where it intersects with slot **212** as best seen in FIG. **12**. Slot **230** is sized and configured to receive a cutting instrument, such as a saw, therein. In some embodiments, as described below, slot **230** is used as a cutting guide for performing a posterior chamfer cut on a bone, such as a talus. A person of ordinary skill in the art will understand that slot **230** may be used for performing other cuts.

(146) Turning now to FIGS. **13-19**, arm component **240** includes a base **242** from which a peg **244** and an arm **246** extend. Peg **244** extends from surface **248** of base **242** and is sized and configured to be received within passageway **224** defined by body **204** of base component **202**. In some embodiments, peg **244** may be received within passageway **224** such that peg **244** can be slid along the length of passageway **224**. Although not shown, one of ordinary skill in the art will understand that peg **244** may include a dovetail or other coupling configuration to mate with a complementary configuration provided by passageway **224** to prevent peg **244** from being easily disengaged from passageway **224**.

(147) In some embodiments, the base **242** and arm **246** may be formed separately and joined together to form arm component **240**. In such embodiments, base **242** may define a cavity **241**, as illustrated in FIGS. **14-16**, sized and configured to receive at least a portion of arm **246**. However, one of ordinary skill in the art will understand that arm component **240** may be a monolith in some embodiments.

(148) Arm **246** of arm component **240** may include a coupling end **250** for coupling arm component **240** to a surgical tool. For example, as shown in FIGS. **17-19**, coupling end may define an opening **252** sized and configured to receive, at least partially, a cutting tool therein. One of ordinary skill in the art will understand that the shape of opening **252**, although illustrated as being generally circular, may vary depending on the type of cutting tool that is to be supported. Opening **252** may communicate with a gap **254** in arm **246**. In some embodiments, arm **246** may include a raised portion **256** forming a stop along a surface **258** of arm **246**. The stop may be configured to engage a tab **270** that extends outwardly from a sleeve **268** as best seen in FIG. **6**. For example, in some embodiments, a portion of the cutting tool **260** may be received within the sleeve **268**, which is then locked into position of arm **246** by way of the tab **270** engaging the stop.

(149) In use, the guide **200** is placed on a bone, such as a flat formed on the talus. For example, bottom surface **210** may be placed on a flat formed on the talus and the position of guide **200**, or at least base component **202**, on the talus may be checked using fluoroscopy. In some embodiments, a fluoroscopic image may be obtained of the sagittal plane such that slot **212** may be seen. As noted above, the angle of the leading edge **232** of slot **212** (FIG. **8**) may identify the angle at which slot **230** extends through body **204** and thus the angle at which a posterior chamfer cut will be made. Once the desired position of the guide **200** on bone has been achieved, the guide **200** may be secured to the bone by inserting fixation elements into holes **220**.

(150) With guide **200** secured to the bone, a chamfer may be formed on the bone by passing a cutting instrument, such as a bone saw, through slot **230**. In some embodiments, the chamfer is formed on a posterior portion of a talar dome.

(151) Another cutting instrument, such as a reamer or the rotating cutting tool **260** illustrated in FIGS. **22-25**, may be used to form another chamfer and/or flat cuts on the bone. For example, the shaft **262** of the cutting tool **260** may be inserted into opening **252** defined by arm component **240**. As noted above, arm component **240** may be slideably coupled to base component **202** via the engagement of peg **244** extending from surface **248** of base **242** that is received within passageway **224** defined by base component **202**. In some embodiments, the shoulder **264** of cutting tool **260** is

positioned adjacent to and/or in abutment with arm **246**. Bony cuts may then be made using the teeth **266** of the cutting tool **260** by moving the arm component **240** linearly along passageway **224** such that the cutting instrument moves in a medial-lateral/lateral-medial direction as will be understood by one of ordinary skill in the art. A dowel, such as dowel **280** shown in FIGS. **20-21**, may be inserted into the hole **238** defined by base component **202** to limit the travel of arm component **240**. In some embodiments, the chamfer formed using the guidance provided by arm component **240** is a chamfer formed on the anterior portion of a talar dome.

(152) Once the cuts have been made using guide **200**, the fixation elements may be removed from their engagement with the guide **200** and bone, and then the guide **200** may be removed from its abutment with the bone. Additional and/or finishing cuts may be made once the guide **200** is removed.

(153) Resection Guide

(154) FIGS. **26-31** illustrate one example of a resection guide **300** in accordance with some embodiments. Resection guide **300** may be used to perform both posterior and anterior chamfer resections. Referring first to FIG. **26**, resection guide **300** has a body extending from an engagement end **304** for coupling to a bone, such as a talus, and a guide end **306**. Engagement end **304** includes a flat surface **308** for being positioned on a flat formed on a bone. Engagement end **304** also defines a slot **310** extending across body **302** (i.e., from first side **312** to second side **314**) and a pair of holes **316-1**, **316-2** (collectively, “holes **316**”). In some embodiments, holes **316** are positioned along opposite sides **312**, **314** of body **302**. Holes **216** are sized and configured to receive a fixation element, such as the threaded section **352** of threaded pins **350** illustrated in FIGS. **32-33**, as described in greater detail below.

(155) As best seen in FIGS. **26**, **29**, and **30**, a slot **318** is formed through body **302** extending from the top side **326** to flat surface **308**. In some embodiments, holes **316** and slot **318** are parallel with each other as they extend through body **302**; however, one of ordinary skill in the art will understand that holes **316** and slot **318** may not be arranged such that they are parallel with one another. Slot **318** is sized and configured to receive a cutting tool, such as a saw, therethrough to provide a guide for making posterior chamfer cuts. As best seen in FIG. **29**, slot **318** aligns and is in communication with slot **310** such that slot **310** may provide a surgeon with a visual indication of the angle and location of the posterior chamfer cut when viewing the guide **300** under sagittal plane fluoroscopy. To facilitate visualization under fluoroscopy, some or all of resection guide **300** may be formed from radiopaque material.

(156) Guide end **306** includes an extension **322** extending from body **302** such that extension **322** has a thickness that is greater than a thickness of body **302** as best seen in FIG. **31**. Extension **322** defines a guide channel **324** and one or more holes **326-1**, **326-2** (collectively, “holes **326**”) each extending from front surface **328** of extension **322** to rear surface **330** of extension **320**. Guide channel **324** is sized and arranged to guide a cutting tool, such as the reamer **370** shown in FIGS. **38-40**, for guiding the cutting tool when making the anterior chamfer and/or flat cuts. In some embodiments, holes **326** are arranged such that they are not parallel with one another (e.g., are divergent) to firmly secure guide **300** to a bone while making the anterior chamfer and/or flat cuts. Guide **300** may define another pair of holes **332-1**, **332-2** (collectively, “holes **332**”) that extend from top side **320** (FIG. **29**) to underneath side **334** (FIG. **30**). In some embodiments, holes **332** are spaced equally to the holes **116** of talar dome trial **100** shown in FIGS. **1-5**.

(157) In use, resection guide **300** may be coupled to a flat formed on a bone, such as a talus, by placing surface **308** on the formed flat. The location of guide **300** relative to the bone may be checked using fluoroscopy. For example, a surgeon or other practitioner may check the position of the guide **300** on a bone to confirm that slot **310**, which identifies the location and angle of slot **318** (and thus the location and angle of a chamfer cut) is at the desired location. When the desired positioning of guide **300** has been achieved, pins, k-wires, or other fixation elements, such as threaded pins **350**, may be inserted through holes **316** to secure guide **300** to the bone. For

example, pins **350** may be inserted by coupling a T-handle (not shown) to the engagement shoulder **356**) and then using the T-handle to tap the self-tapping threaded end **352** into the bone. In some embodiments, the threaded end **352** is threaded into bone until shoulder **354** comes into contact with resection guide **300**. As shown in FIGS. **32** and **33**, the engagement shoulder **356** may be disposed between unthreaded sections **358-1**, **358-2** (collectively, “unthreaded sections **358**” or “unthreaded section **358**”). Additional securement may be provided by inserting other fixation devices through holes **326**.

(158) With guide **300** secured to the bone, bony cuts may be made using the guide **300** to form posterior and/or anterior chamfers. For example, making posterior chamfer cuts may include inserting a saw into and through slot **318**. The saw may be moved side-to-side within and along slot **318**. An anterior chamfer and/or anterior flat may be formed by inserting another cutting tool, such as an angled reamer **380** illustrated in FIGS. **38-40**, into guide channel **324**. As best seen in FIG. **41**, the reamers may be moved along guide channel **324** to form the anterior chamfer and/or flat cuts as will be understood by one of ordinary skill in the art.

(159) Once the desired cuts have been made using guide **300**, the fixation elements may be removed from their engagement with the bone and guide **300** such that guide **300** may also be removed from its contact with bone. In some embodiments, with guide **300** removed, additional finishing cuts may be made to the posterior and/or anterior chamfers as will be understood by one of ordinary skill in the art.

(160) Anterior Cut Guide

(161) FIGS. **42-47** illustrate one example of an anterior cutting guide in accordance with some embodiments. Cutting guide **400** includes an upper surface **402** and a lower surface **404**. In some embodiments, upper surface **402** includes a first planar surface **402-1** and a second planar surface **402-2** that is disposed at angle relative to first planar surface **402-1**. Lower surface **404** may include a first planar surface **404-1**, a second planar surface **404-2**, a third planar surface **404-3**, and a fourth planar surface **404-4**. In some embodiments, planar surfaces **404-1**, **404-3** are parallel to one another and planar surfaces **404-2**, **404-4** are disposed at angles with respect to surfaces **404-1**, **404-3**.

(162) As best seen in FIGS. **45**, planar surface **404-1** of lower surface **404** includes one or more pegs **406-1**, **406-2** (collectively, “pegs **406**”) extending therefrom. In some embodiments, pegs **406** are sized and arranged (e.g., spaced apart) such that cutting guide **400** may be coupled to a base component, such as the talar resection guide base **2100** disclosed in the '561 Patent. For example, pegs **406** may be received within holes **2116**, **2118** defined by the resection guide base **2100** as described in greater detail below.

(163) Guide **400** defines a pair of slots **408**, **410** extending parallel to one another in a lengthwise direction across guide **400**. Slot **408** extends from planar surface **402-2** of upper surface **402** to planar surfaces **404-2**, **404-3** of bottom surface **404**. Slot **410** extends from planar surface **402-1** of upper surface **402** to planar surfaces **404-2**, **404-3** of bottom surface **404**.

(164) In some embodiments, slot **408** includes an enlarged area **412** disposed along its length. As shown in FIGS. **42**, **44**, and **45**, enlarged area **412** may have a circular geometry, although one of ordinary skill in the art will understand that enlarged area **412** may have other geometries or shapes. Slot **410** may also include an enlarged area **414** along its length. Like enlarged area **412**, enlarged area **414** may have a circular shape, but may have other shapes as will be understood by one of ordinary skill in the art.

(165) In use, the guide **400** may be coupled to a base component, such as the talar resection guide base **2100** disclosed in the '561 Patent. As disclosed in the '561 Patent, the talar guide base **2100** may be coupled to a bone, such as a talus, by a fixation element, such as a fixation pin or k-wire. Coupling the guide **400** to the base component may include inserting pegs **406** into holes **2116**, **2118**. With guide **400** coupled to the base component, slot **408** is aligned with slot **2102** defined by the base component. An initial plunge cut may be made by inserting a cutting tool, such as a

reamer, into enlarged area **412** along the length of slot **408**. The same or another cutting tool may then be moved along the length of slot **408** to cut the bone.

(166) Once the desired cutting has been achieved using the guidance provided by **408**, guide **400** remains coupled to the base component, slot **410** is aligned with slot **2124** defined by the base component. An initial plunge cut may be made by inserting a cutting tool, such as a reamer, into enlarged area **414** along the length of slot **410**. Once the desired cutting has been achieved using the guidance provided by **408** and **410**, guide **400** may be removed from its engagement with the base component. The same or another cutting tool may then be moved along the length of slot **410** to make additional bone resections as will be understood by one of ordinary skill in the art.

(167) Posterior Referencing Chamfer Guides

(168) FIGS. **48-64** illustrate examples of posterior-referencing chamfer cutting guides in accordance with some embodiments. Referring first to FIGS. **48-54**, a posterior chamfer cutting guide **500** is shown having a body **502** defining a cutting slot **504**. In some embodiments, slot **504** includes one or more protection holes **506** along its length. Slot **504** (and holes **506**) extends at an angle from rear side **508** to the bottom side **510** as best seen in FIG. **51**.

(169) Body **502** defines a pair of holes **512-1**, **512-2** (collectively, “holes”), with hole **512-1** being positioned alongside **514** and hole **512-2** being positioned alongside **516**. Body **502** also includes one or more tangs **518-1**, **518-2** (collectively, “tangs **518**”) extending away from the rear side **508**. Each tang **518-1**, **518-2** defines a respective hole **520-1**, **520-2** (collectively, “holes **520**”) sized and configured to receive a fixation element, such as a k-wire or pin, therein.

(170) As shown in FIG. **54**, the cutting guide **500** is placed on a bone, such as a talus, and is secured to the bone by inserting one or more fixation elements through holes **512** and/or holes **520**. Additional fixation elements may be inserted through the protection holes **506** disposed along the length of slot **504**. A cutting tool, such as a bone saw, is inserted through slot **504** to make a chamfer cut, such as a posterior chamfer cut. The cutting guide **500** may then be removed from its engagement with the fixation elements and another guide may be installed to make an anterior chamfer cut.

(171) One example of such a cutting guide is shown in FIGS. **55-64**. Guide **600** includes a base component **602** that is sized and configured to support an arm component **620** such that arm component **620** may be moved relative to base component **602**. Base component **602** extends from a leading end **604** to a trailing end **606**. Base component **602** defines one or more holes **608-1**, **608-2** (collectively, “holes **608**”) each being sized and configured to receive a fixation element, such as a pin or k-wire, therethrough for securing the base component **602** to a bone as illustrated in FIG. **55**. Trailing end **606** includes a beam **610** extending upwardly away from leading end **604** of base component **602**. In some embodiments, beam **610** includes one or more stops **612-1**, **612-2** (collectively, “stops **612**”) extending from planar surface **611** of beam **610** and includes one or more tenons or projections **614-1**, **614-2** (collectively, “projections **614**”) as best seen in FIG. **60**. Beam **610** may also include a cutout **616** sized and configured to receive arm component **620** as described below. Base component may also define a hole **618** for additional fixation into the bone.

(172) As best seen in FIGS. **61-64**, arm component **620** extends from a coupling end **622** to another end **624**. Coupling end **622** defines a mortise or recess **626** sized and configured to engage beam **610** and projections **614**. For example, in some embodiments, the combination of projections **614** and recess **626** form a dovetail or tenon-mortise connection or joint that enables arm component **620** to move, such as by sliding, relative to base component **602** along beam **610** and projections **614**. In some embodiments, arm component **620** includes a pair of opposed tapers **628-1**, **628-2** (collectively, “tapers **628**”) along its length between coupling end **622** and end **624**. Tapers **628** may be provided to facilitate gripping or enhanced ergonomics when arm component **620** is manipulated by a physician or other practitioner or user.

(173) A guide hole **630** extends from a first side **632** of arm component **620** to a second side **634** of arm component **620**. As best seen in FIG. **55**, guide hole **630** is arranged such that a central axis

defined by guide hole **630** is arranged perpendicularly with respect to a longitudinal axis defined by beam **610**. Guide hole **630** is sized and configured to receive a shaft of a cutting tool, such as a reamer.

(174) In use, guide **600** is placed on a bone as shown in FIG. **55**. For example, in some embodiments, the base component **602** of guide **600** is placed on a flat formed on a superior portion of a talus. One or more fixation elements, such as k-wires or pins, are inserted through holes **608** to secure the base component to the bone. The arm component **620** may be coupled to the base component **602** by aligning the coupling end **622** of arm component **620** with cutout **616** defined by beam **610**. When aligned, the dovetail/tenon-mortise connection between arm component **620** and base component **602** may be formed as will be understood by one of ordinary skill in the art.

(175) With the guide **600** secured to bone, bony cuts may be made to form a chamfer or chamfers. For example, a cutting instrument, such as the angled cutting instrument **650** illustrated in FIGS. **65-67**, is coupled to the guide **600** by inserting a shaft **652** of the cutting instrument **650** into the guide hole **630** defined by arm component **620**. The cutting instrument may be secured to the arm component via a nut, such as nut **670** illustrated in FIGS. **68-70**, by threading the nut **670** onto threaded section **654** of shaft **652** of cutting instruction **650**. The cutting tool is guided through making bony cuts as the surgeon or practitioner may grasp the arm component **620** via the tapers **628** and move the arm component **620** relative to the base component **602** by sliding arm component **620** along beam **610** from stop **612-1** to stop **612-2**.

(176) In some embodiments, the arm component **620** of guide **600** may be swapped out for another arm component, such as arm component **620A** shown in FIGS. **71-75**, to perform the anterior chamfer and flat cuts on the talus instead of arm component **620**. As shown FIGS. **71-75**, arm component **620A** extends from a coupling end **622A** to another end **624A**. Coupling end **622A** defines a mortise or recess **626A** sized and configured to engage beam **610** and projections **614**. For example, in some embodiments, the combination of projections **614** and recess **626A** form a dovetail or tenon-mortise connection or joint that enables arm component **620A** to move, such as by sliding, relative to base component **602** along beam **610** and projections **614**. In some embodiments, arm component **620A** includes a pair of opposed tapers **628A-1**, **628A-2** (collectively, “tapers **628A**”) along its length between coupling end **622A** and end **624A**. Tapers **628A** may be provided to facilitate gripping or enhanced ergonomics when arm component **620A** is manipulated by a physician or other practitioner or user.

(177) A guide hole **630A** extends from a first side **632A** of arm component **620A** to a second side **634A** of arm component **620A**. Guide hole **630A** is arranged such that a central axis defined by guide hole **630A** is arranged perpendicularly with respect to a longitudinal axis defined by beam **610**. A second guide hole **636A** is defined by arm component **620A**. Guide hole **636A** is disposed at an angle relative to guide hole **630A**. As described in greater detail below, the angle at which guide hole **636A** is oriented relative to guide hole **630A** facilitates making the anterior flat cut. Both guide holes **630A** and **636A** are sized and configured to receive a shaft of a cutting tool, such as reamer **680** illustrated in FIGS. **76-79**.

(178) Arm component **620** may be removed by uncoupling the cutting tool, e.g., cutting tool **650**, from its engagement with arm component **620**, which may include unscrewing nut **670**. With cutting tool **650** uncoupled from arm component **620**, arm component **620** may be uncoupled from base component **602** by aligning arm component **620** with the cutout **616** of beam **610** and rotating arm component **620** such that it disengages beam **610**. In some embodiments, arm component **620A** may be coupled to base component **602** instead of arm component **620** by aligning coupling end **622A** with cutout **616** of beam **610** and engaging the coupling end **622A** with projections **614** and beam **610**.

(179) A cutting tool, such as reamer **680**, may then be inserted into hole **630A** such that a first cut may be performed. The reamer **680** may be guided in making the first cut by sliding arm **620A**

along beam **610** from stop **612-1** to stop **612-2** as illustrated in FIG. **80**. The reamer **680** may then be removed from hole **630A** and inserted into hole **636A** to make a second cut as illustrated in FIG. **81**.

(180) Once the cuts have been made using guide **600**, the fixation elements may be removed from their engagement with the guide **600** and bone, and then the guide **600** may be removed from its contact with the bone.

(181) Anterior Referencing Chamfer Guides

(182) FIGS. **82-107** illustrate a series of guides that may be used with anterior referencing to make both anterior chamfer cuts and posterior chamfer cuts. Referring first to FIGS. **82-87**, guide base **702** includes a planar bottom surface **704** for mating to a flat formed on a bone and defines one or more holes **708-1**, **708-2** (collectively, “holes **708**”). Upper portion **706** of base **702** includes a beam **710**, which may be similar to beam **610** described above in that beam **710** may include stops **712-1**, **712-2** (collectively, “stops **712**”), projections **714-1**, **714-2** (collectively, “projections **714**”), and a cutout **716**.

(183) Base **702** defines a hole **718** extending from upper surface **720** to lower surface **704**. Surface **722** of base **702** defines a central hole **724** and two adjacent holes **726-1**, **726-2** (collectively, “holes **726**”). Holes **726** are sized and configured to receive dowels as will be described in greater detail below. A slot **728** is defined by base **702** and extends at an angle from surface **722** to bottom surface **704** as best seen in FIGS. **82** and **87**.

(184) FIGS. **88-93** illustrate one example of a locator guide **800** that may be used to position guide base **702**. Locator guide **800** includes a body **802** including a pair of spaced apart legs **804-1**, **804-2** (collectively, “legs **804**”). Each leg **804-1**, **804-2** defines a respective hole **806-1**, **806-2** (collectively, “holes **806**”) each sized and configured to receive a fixation element, such as a k-wire or pin, for securing the locator guide **800** to bone.

(185) Bottom surface **806** of body **802** may be planar to facilitate positioning locator guide **800** on a flat formed on a bone, such as a talus. Body **802** defines a through hole **808** extending from surface **810** disposed between legs **804** to rear surface **812**. Hole **808** is sized and configured to receive an adjustment bolt **814** therein in a threaded engagement as will be understood by one of ordinary skill in the art. As best seen in FIG. **92**, rear surface **812** also defines a pair of blind holes **816-1**, **816-2** (collectively, “holes **816**” or “blind holes **816**”) that extend inwardly from surface **812** and are sized and configured to receive a respective dowel pin **818** therein. In some embodiments, body **802** may define a hole **820** extending from top surface **822** to bottom surface **806**. Hole **820** is sized and configured to receive a dowel pin **824** therein.

(186) Locator guide **800** may be used to position guide base **702**, which is shown in FIG. **84**. More particularly, dowels **818** extending from guide **800** may be inserted into holes **726** defined by guide base **702**. With dowels **818** positioned within holes **726**, hole **808** is aligned with hole **724** and bolt **814** is used to assemble locator guide **800** to guide base **702**. Fixation elements may be inserted into holes **806** defined by legs **804** of locator guide **800**.

(187) With guide base **702** properly positioned, fixation elements may be inserted into the holes **708** defined by guide base **702** to secure guide base to the bone. The locator guide **800** may then be removed from its engagement with guide base **702** and the bone as will be understood by one of ordinary skill in the art. An arm component, such as arm component **620** or arm component **620A**, may then be coupled to guide base **702** in the same manner to facilitate making anterior chamfer cuts as described above with respect to cutting guide **600**.

(188) Guide base **702** may also be used in making posterior chamfer cuts. For example, FIGS. **94-99** illustrate one example of a posterior cutting guide **900** that may be used with guide base **702** to form a posterior chamfer cut. Referring first to FIG. **94**, cutting guide **900** includes a body **902** defining one or more holes **904-1**, **904-2** (collectively, “holes **904**”) for receiving fixation elements, such as k-wires or pins, to secure the cutting guide **900** to bone. As best seen in FIG. **95**, the bottom surface **906** of cutting guide **900** may be angled to mate to a previously formed anterior chamfer.

(189) Body **902** also defines a slot **908** extending from surface **910** to side **912**. Side **912** also defines one or more holes **914-1**, **914-2** (collectively, “holes **914**”) as shown in FIG. **96**. Holes **914** may be blind holes extending inwardly from side **912** and being sized and configured to receive a dowel pin **916** as illustrated in FIGS. **94** and **95**. Holes **914** are positioned on body **902** such that they align with holes **726** defined by guide base **702**. Similarly, slot **908** extends through body **902** such that it aligns to slot **728** when dowels **916** are received within both **914** and holes **726**.

(190) FIG. **100** illustrates one example of guide base **702** being coupled to bone and cutting guide **900** being coupled to bone and to guide base **702**. A posterior chamfer cut may be made by inserting a cutting tool, such as a bone saw or chisel, into and through slot **908** defined by cutting guide **900** and into and through slot **728** defined by guide base **702**.

(191) Another example of a posterior chamfer cutting guide is illustrated in FIGS. **101-106**. Cutting guide **900A** includes a body **902A** defining at least one hole **904A-1**, **904A-2** (collectively, “holes **904**”) sized and configured to receive a fixation element for securing the cutting guide **900A** to bone. Like cutting guide **900**, cutting guide **900A** includes a bottom surface **906A** that is angled to mate to a previously formed anterior chamfer. Body **902A** also defines a slot **908A** that extends from surface **910A** to surface **912A**.

(192) Guide **900A** differs from guide **900** in that guide **900A** has a different coupling mechanism than guide **900**. More particularly, guide **900A** includes an extension arm **920A** having projections **922A-1**, **922A-2** (collectively, “projections **922A**”) that are configured to engage projections **714** of guide base **702** to couple guide **900A** to guide base **700** as shown in FIG. **107**. When guide **900A** is coupled to guide base **702**, slot **908A** of guide **900A** is aligned with slot **728** of guide base **702** such that a cutting tool, such as a bone saw or chisel, may be inserted into and through slots **908A** and **728** to form a posterior chamfer on the bone.

(193) Additional Guides (FIGS. **108-133**)

(194) FIGS. **108-113** provide various views of a cutting guide in accordance with some embodiments. Cutting guide **1000** may be placed on a flat formed on a bone, such as a talus, in some embodiments. More particularly, cutting guide **1000** includes a body **1002** having a bottom surface or side **1004** that may be placed on a flat formed on a bone. Body **1002** also includes a pair of legs **1006-1**, **1006-2** (collectively, “legs **1006**”) each defining a first hole **1008-1**, **1008-2** (collectively, “holes **1008**”) and a second hole **1010-1**, **1010-2** (collectively, “holes **1010**”). In some embodiments, holes **1008** and **1010** are formed at an angle relative to one another. For example, holes **1008** may define an axis that is perpendicular to a plane defined by surface **1004** and holes **1010** may extend through legs **1006** at an oblique angle relative to the plane defined by surface **1004**.

(195) Body **1002** also defines an enlarged opening **1012** and a slot **1014**. Both opening **1012** and slot **1014** may extend through the entirety of body **1002**. Opening **1012** is sized and configured to receive a cutting tool, such as a burr, end mill, or other rotating cutting tool, therethrough. Slot **1014** also is dimensioned to receive a cutting tool, such as a saw blade, therein. Body **1002** includes a mating portion **1016** that defines a hole **1018** as best seen in FIGS. **108** and **109**. A pair of mounting holes **1020-1**, **1020-2** (collectively, “mounting holes **1020**”) are defined adjacent to opening **1014** as best seen in FIG. **111**. Lateral holes **1022-1**, **1022-2** (collectively, “holes **1022**” or “lateral holes **1022**”) may also be provided for receiving a fixation element, such as a pin or k-wire, therein to secure the cutting guide to a bone.

(196) FIGS. **114-119** illustrate one example of a cutting guide **1100** that may be coupled to cutting guide **1000**. Guide **1100** has a body **1102** with a leading side **1104** and a trailing side **1106** and defines a pair of enlarged openings **1108**, **1110**. Openings **1108**, **1110** are sized and configured to align with opening **1012** defined by the body **1002** of guide **1000** when guides **1000**, **1100** are coupled to one another as shown in FIG. **120**.

(197) Leading side **1104** may include one or more feet **1112-1**, **1112-2** (collectively, “feet **1112**”) extending therefrom. Each foot **1112-1**, **1112-2** may include a respective coupling member **1114-1**,

1114-2 (collectively, “coupling members **1114**”), as best seen in FIGS. **115**, **116**, and **119**, which extend from the foot and are sized and configured to be received in a respective mounting hole **1020-1**, **1020-2**.

(198) Trailing side **1106** defines a hole **1116** that is sized and configured to align with hole **1018** defined by mating portion **1016** of guide **1000**. For example, hole **1116** defined by guide **1100** should align with hole **1018** when coupling members **1114** are received within mounting holes **1020** and guides **1000**, **1100** are coupled together as shown in FIG. **120**.

(199) FIGS. **121-126** illustrate an example of a base component in accordance with some embodiments. Base component **1200** includes a body **1202** having a bottom side or surface **1204** that is configured to be disposed on a surface of a bone, such as a flat formed on a talus. Body **1202** defines an opening **1206** that extends through body **1202** and is sized and configured to receive a rotating cutting tool therein. A slot **1208** also is defined by body **1202** and is positioned within opening **1206** and extends toward the front side **1210**.

(200) Body **1202** also includes a rear flange **1212** that may define a first pair of holes **1214-1**, **1214-2** (collectively, “holes **1214**”) and a second pair of holes **1216-1**, **1216-2** (collectively, “holes **1216**”). A third pair of holes **1218-1**, **1218-2** (collectively, “holes **1218**”) may also be defined along flange **1212**. Body **1202** may define additional holes **1220-1**, **1220-2**, **1220-3** (collectively, “holes **1220**”) adjacent to opening **1206** and lateral holes **1222-1**, **1222-2** (collectively, “holes **1222**” or “lateral holes **1222**”).

(201) FIGS. **127-133** illustrate one example of a cutting guide **1300** that may be coupled to and used with base **1200**. Guide **1300** includes a body **1302** defining an opening **1304** that is sized and configured to receive a rotating cutting tool therein. Body **1302** may include one or more legs **1306-1**, **1306-2** (collectively, “legs **1306**”) that extend from a lower surface **1308** of body **1302**. Legs **1306** are sized and configured to be received within holes **1214** and **1220** defined by base **1200**. When guide **1300** is coupled to base **1200** with legs **1306** received within holes **1214** or holes **1220**, the opening **1304** defined by guide **1300** aligns with opening **1206** defined by base **1200**. The alignment of openings **1206** and **1304** provides guidance for a rotary cutting tool to make plunge cuts as will be understood by one of ordinary skill in the art.

(202) Although the invention has been described in terms of exemplary embodiments, it is not limited thereto. Rather, the appended claims should be construed broadly, to include other variants and embodiments of the invention, which may be made by those skilled in the art without departing from the scope and range of equivalents of the invention.

Claims

1. A tool, comprising: a base component having a first surface configured to engage a bone, a second surface disposed on an opposite side of the base component with respect to the first surface, and a third surface extending between the first surface and the second surface, the first surface defining a first slot that extends inwardly from a first side of the base component to a second side of the base component, the base component defining a second slot extending inwardly from the third surface and being in communication with the first slot and configured such that a chamfer is formed on the bone by passing a cutting instrument inwardly through the second slot and into the bone, and the second surface defining a passageway; and an arm component, the arm component having a base from which a peg extends, the peg size and configured to be at least partially received within the passageway defined by the base component for coupling the arm component to the base component, wherein the arm component defines an opening sized and configured to receive a first cutting tool therein.

2. The tool of claim 1, wherein the peg of the arm component is configured to slide along a length of the passageway defined by the base component.

3. The tool of claim 1, wherein a width of the opening is greater than a width of a gap in a coupling

end of the arm component that defines the opening.

4. A tool, comprising: a base component extending from a first end to a second end, the base component defining at least one hole adjacent to the first end and including a beam disposed at the second end, the beam including at least one projection extending along its entire length and projecting perpendicular to its length; and an arm component for coupling to the base component, the arm component including a coupling end defining a recess for receiving at least a portion of the beam and the at least one projection to couple the arm component to the base component, the arm component defining at least one hole sized and configured to receive a cutting tool therethrough.

5. The tool of claim 4, wherein the hole defines an axis that is parallel to a plane defined by the beam when the arm component is coupled to the base component.

6. The tool of claim 5, wherein the arm component defines a second hole, the second hole defining an axis that is transverse to the plane defined by the beam when the arm component is coupled to the base component.

7. The tool of claim 4, wherein the base component includes at least one stop positioned along a length of the beam.
